

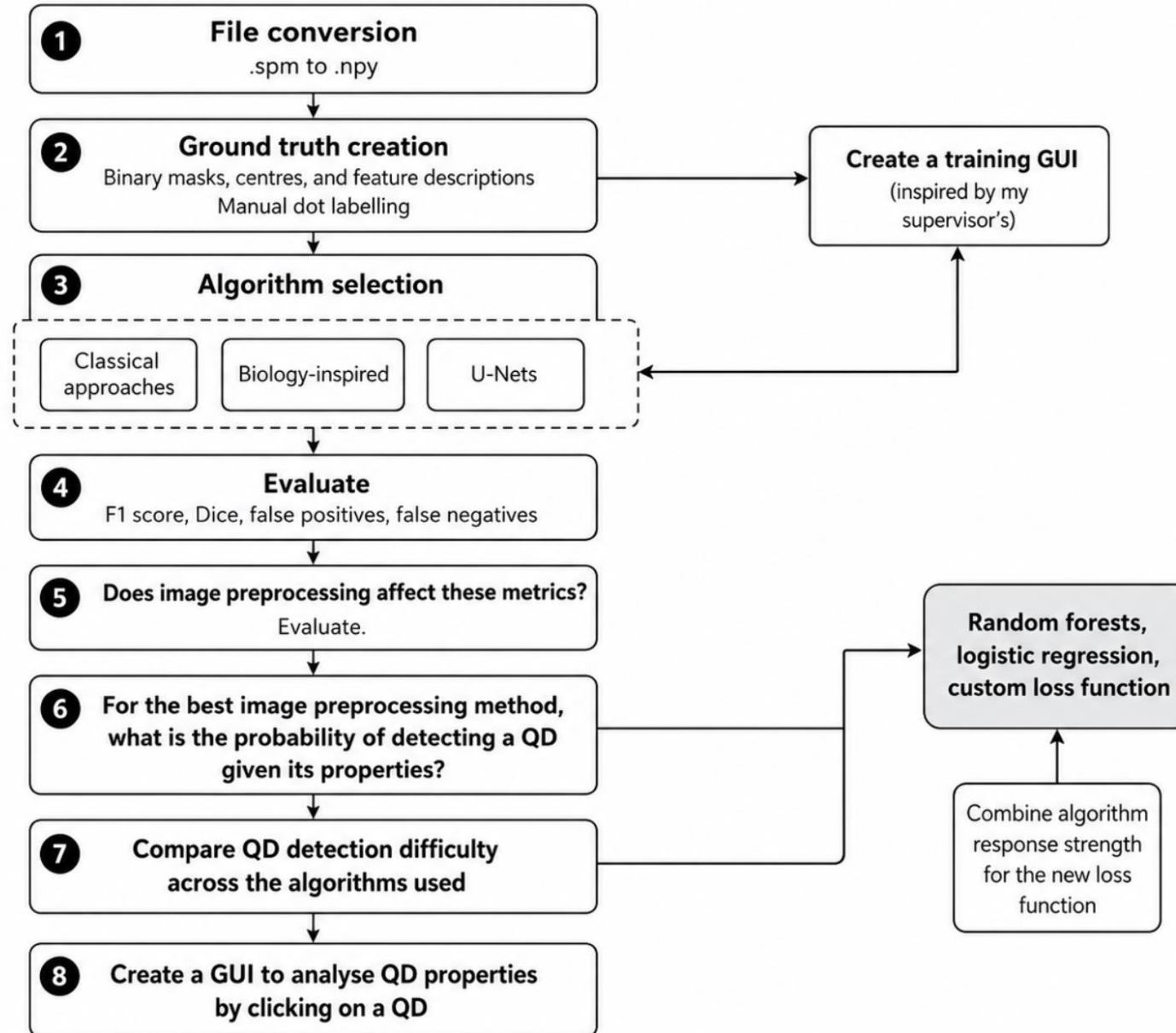
Automating GaN Quantum Dot Detection and Analysis

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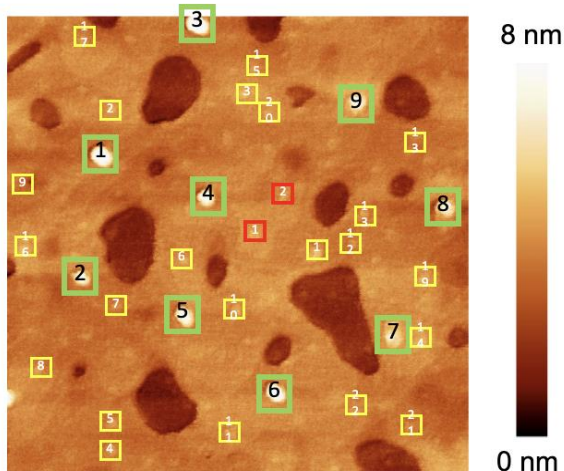
Pipeline



- **The Setup:** 10 classical CV + 7 deep learning models benchmarked.
- **Ground Truth:** 10 manual scans, ~1,800 labelled dots. Augmented and folded to expand the training dataset.
- **Analysis:** looked at why certain dots are missed by algorithms (and why algorithms miss them) to create custom models.
- **Tool:** Custom-built annotation GUI + app to quickly label the dots using the final best model (currently a u-net)

Ground-Truth

CTM 0.35°



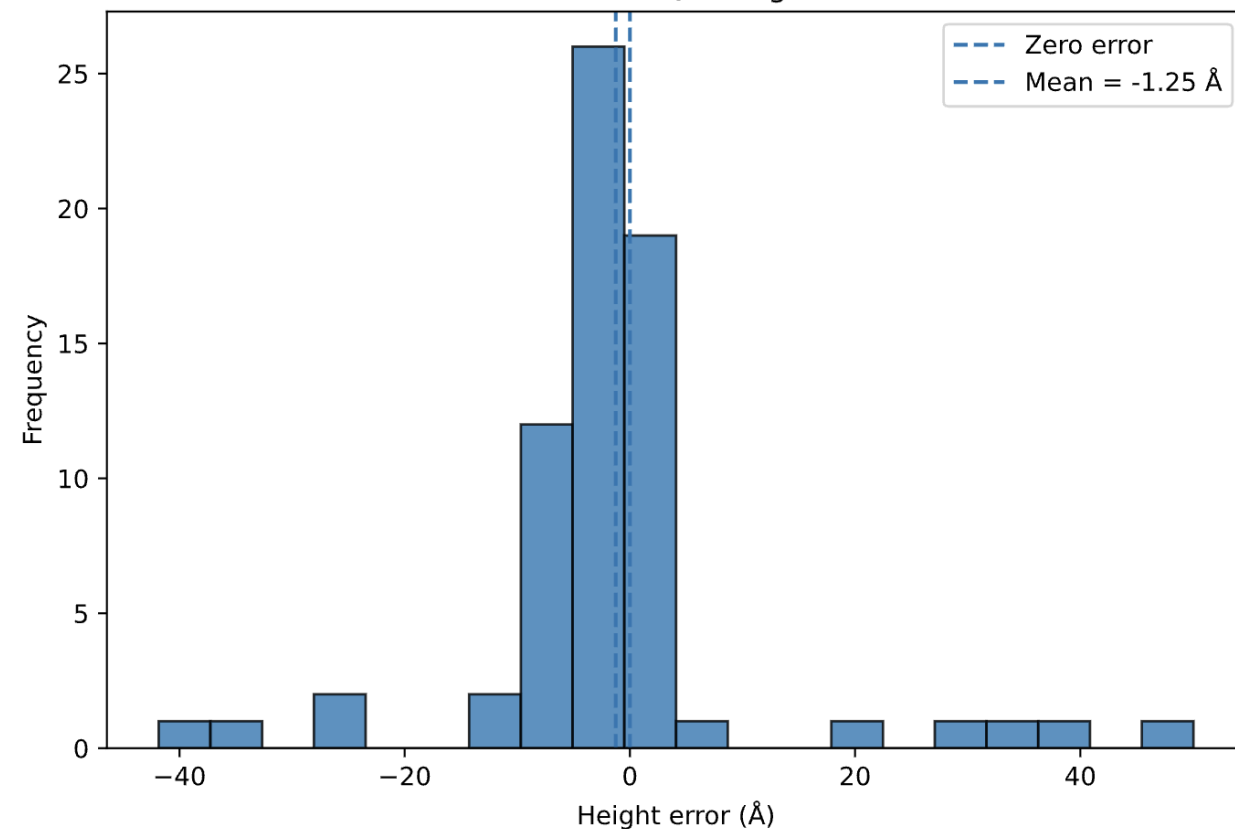
← Xin's data, with labelled height and FWHM

▼ Scan1

- channel_00__0_data_false_negatives.csv
- channel_00__0_data_false_positives.csv
- channel_00__0_data_features.csv
- channel_00__0_data_mask.npy
- channel_00__0_data.npy
- channel_01__1_data.npy
- channel_02__2_data.npy

Ground truth dataset structure updated

Distribution of QD Height Errors

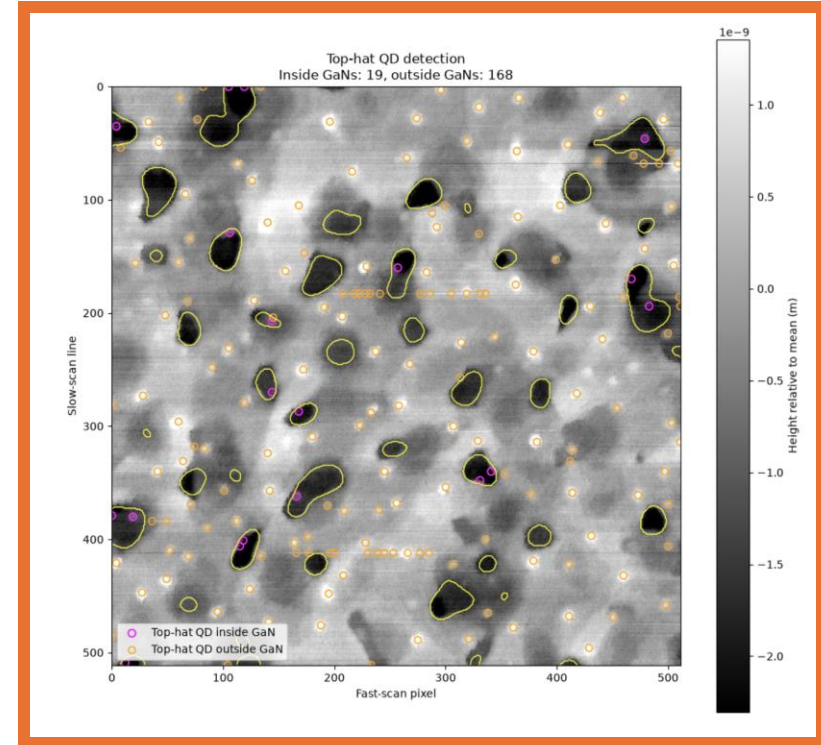
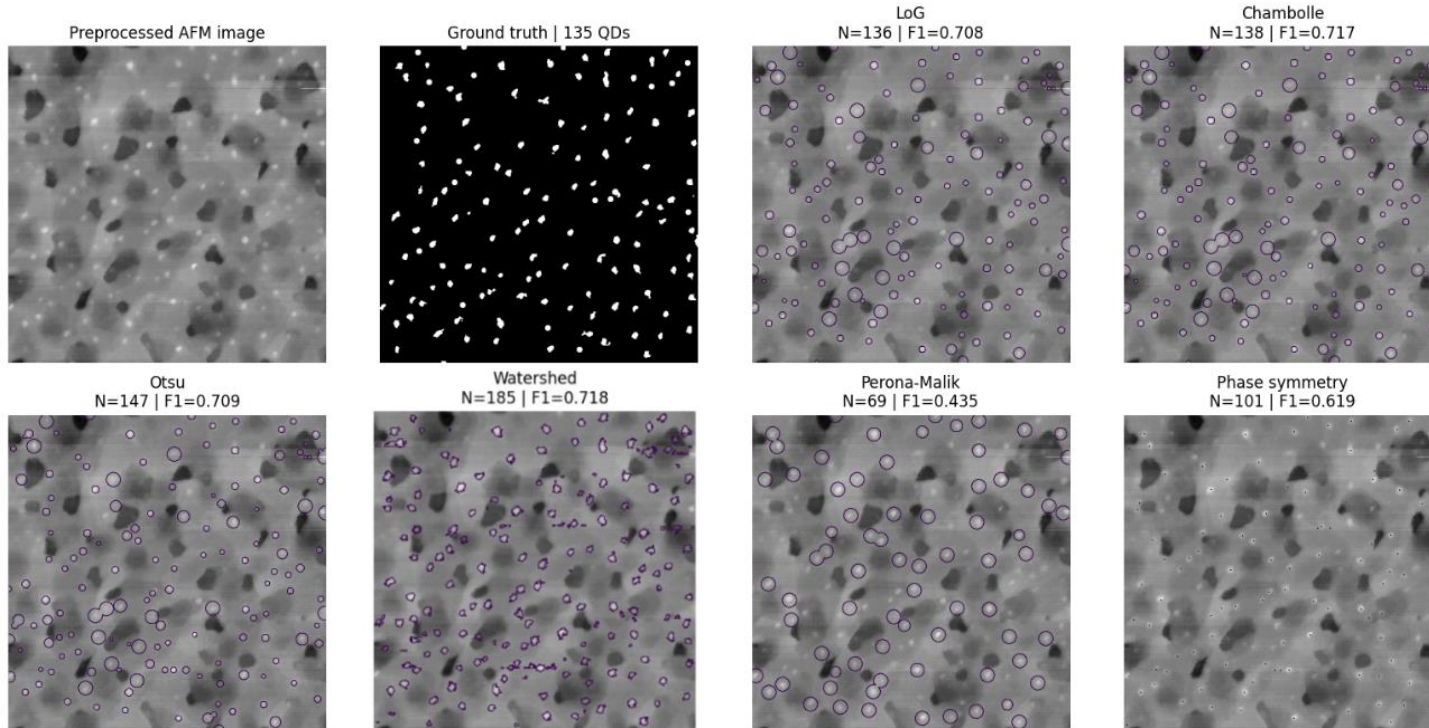


1. How to detect a quantum dot?

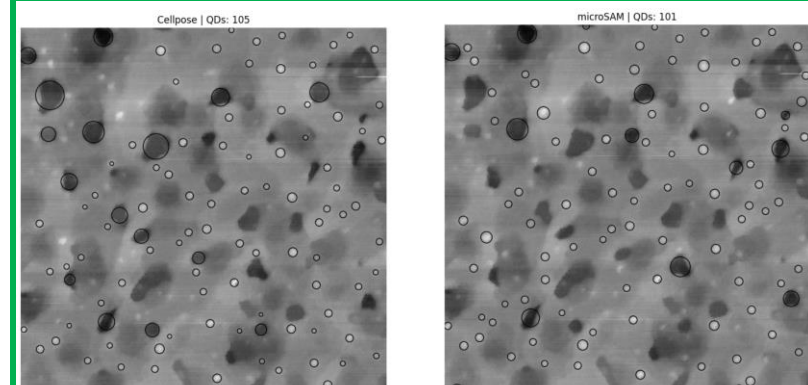
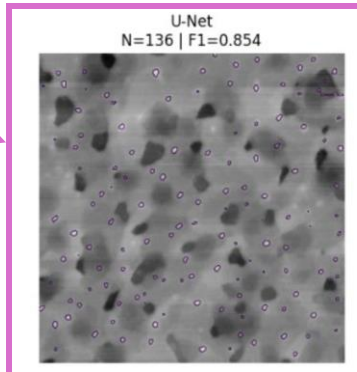
Best performing Classical algorithms



Final result:

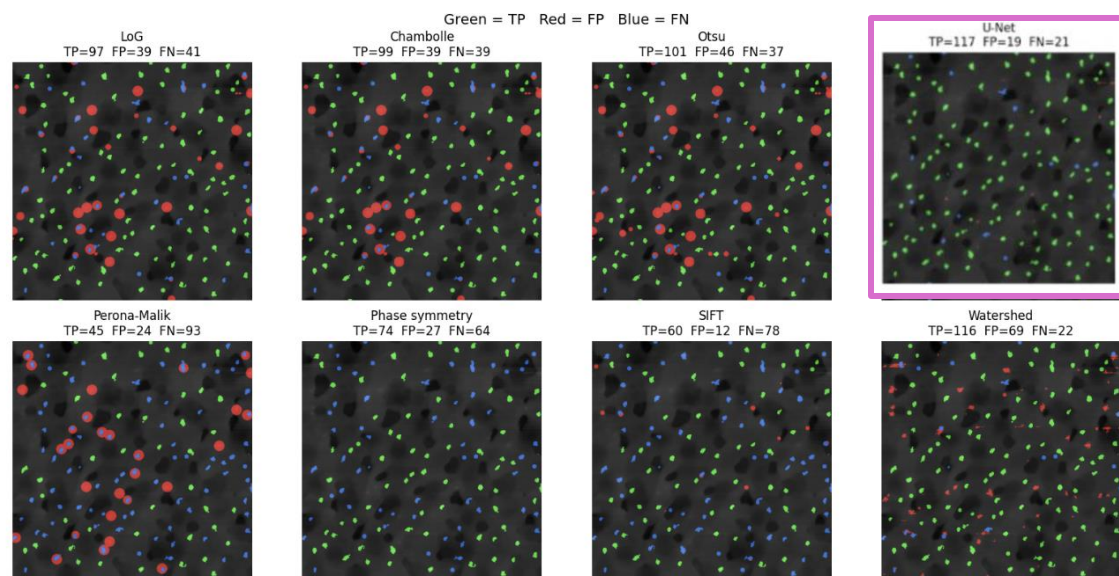


ML Approaches



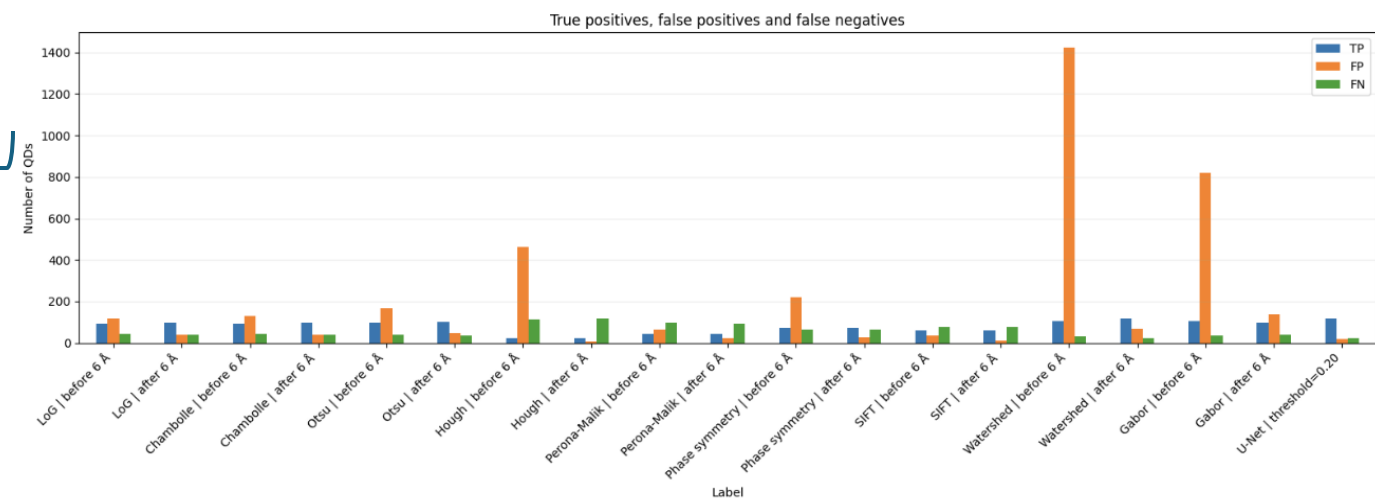
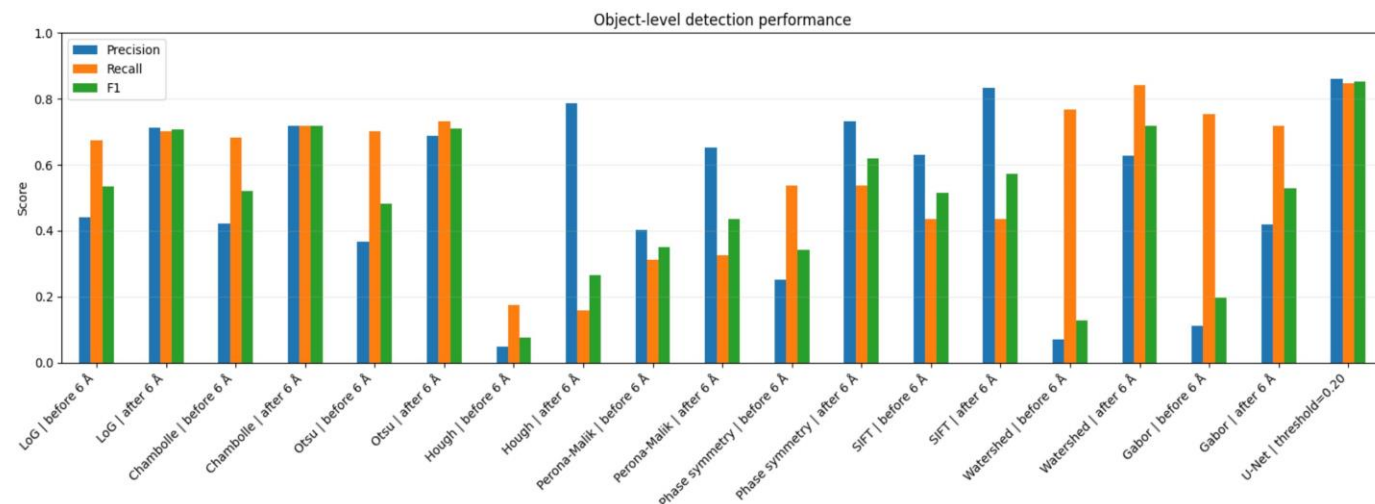
Biology-
Adjacent ML
approaches

2. Evaluate Algorithm Performance

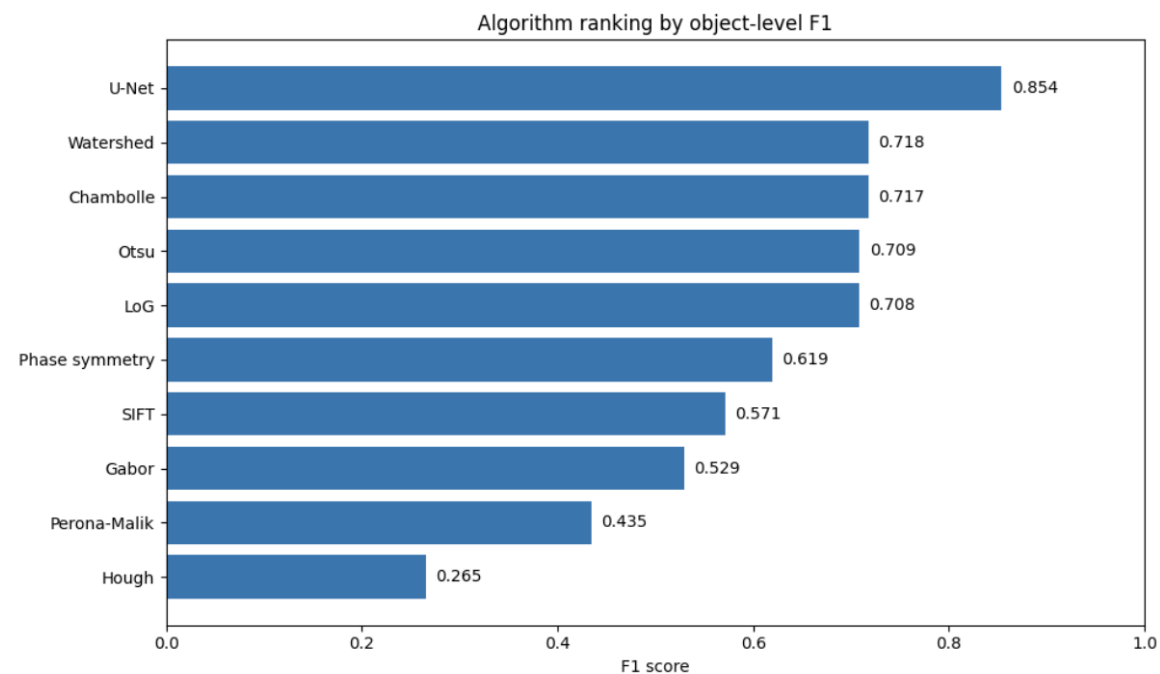


Overlaying results of the best classical algorithms

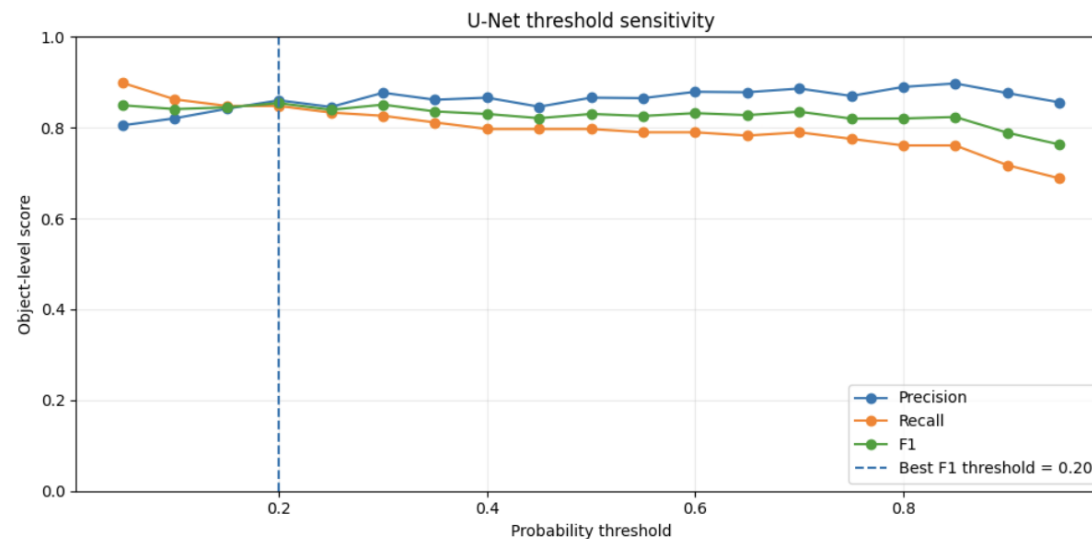
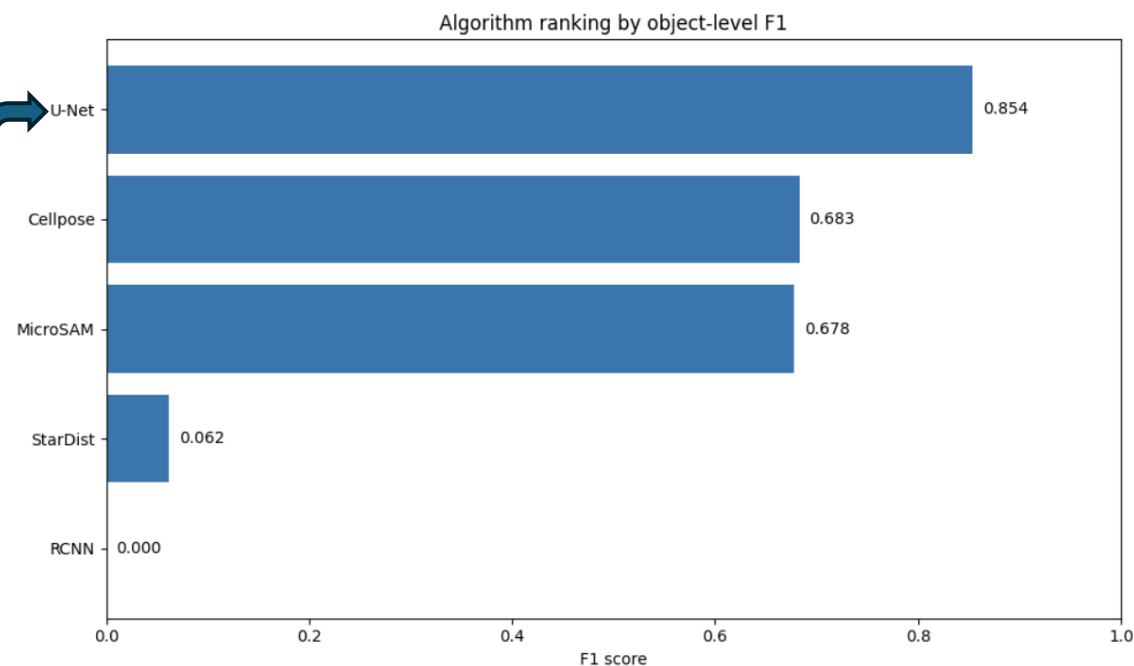
***we tried more, and those came out with the highest F1 score.**



Classical algorithm ranking

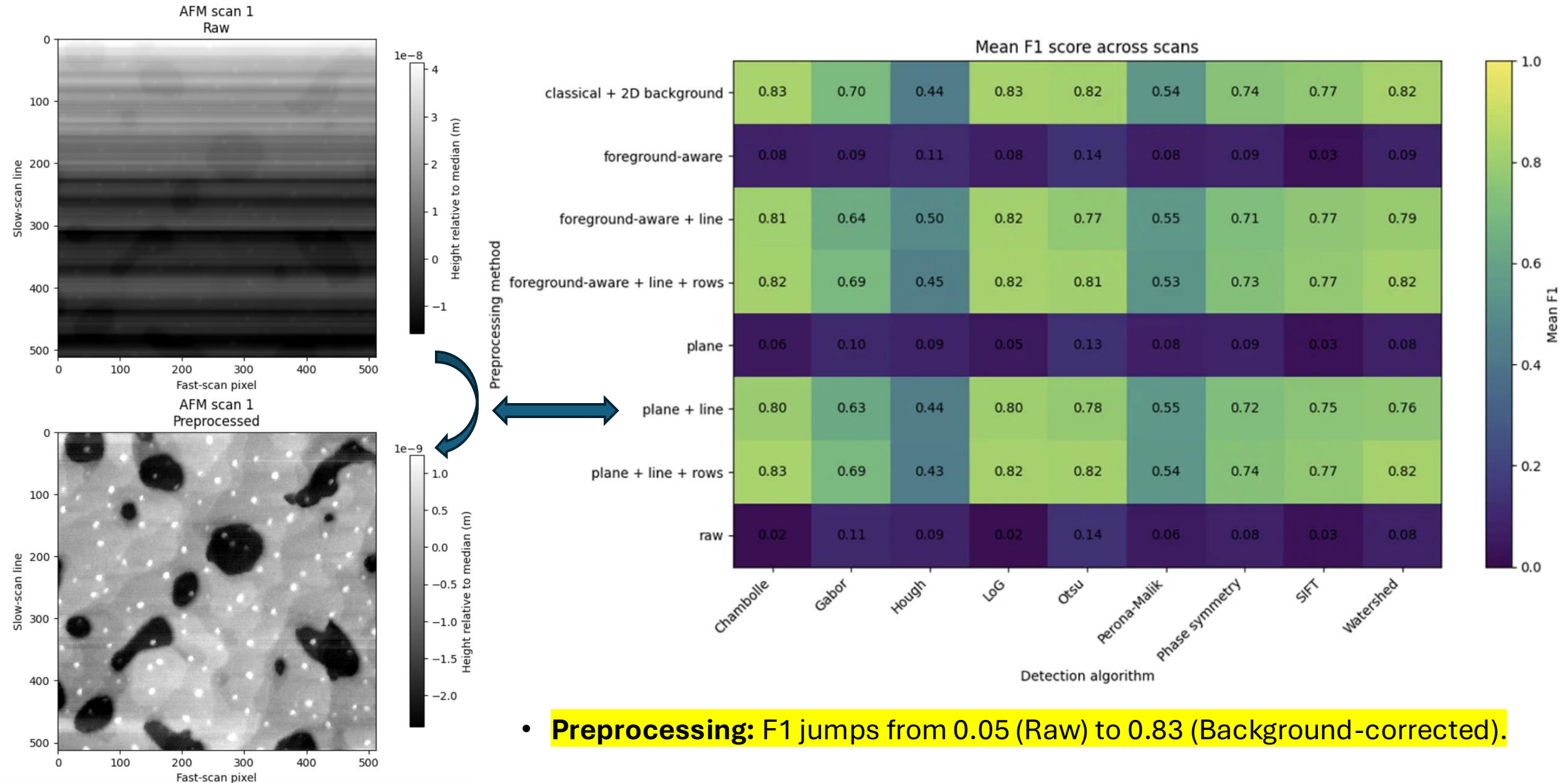


Biology-inspired algorithm ranking

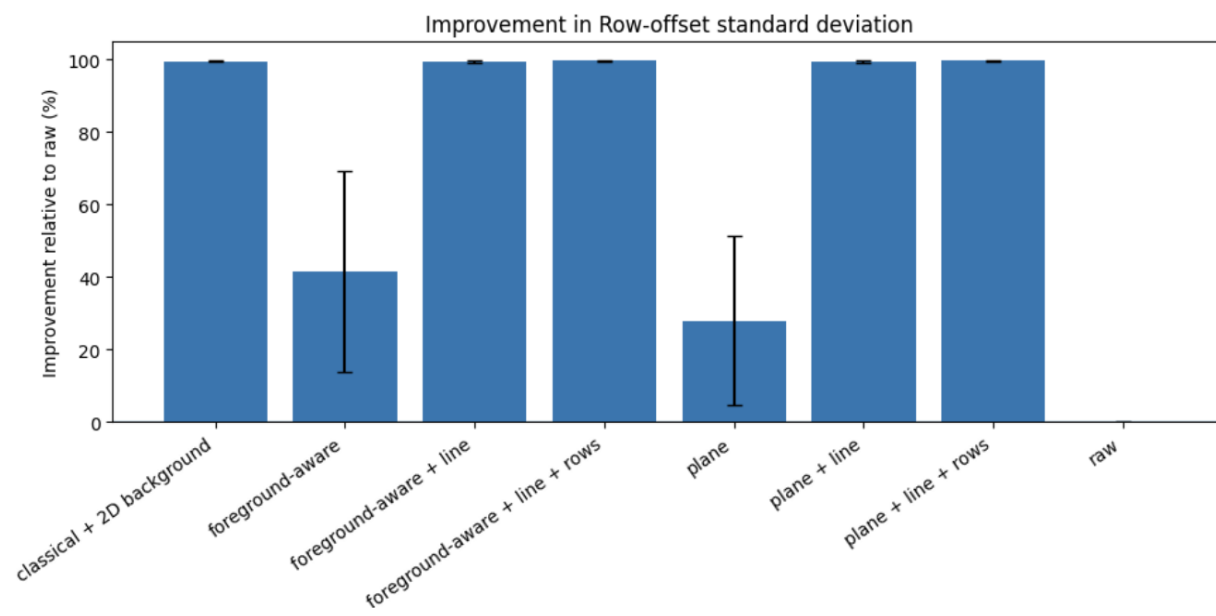
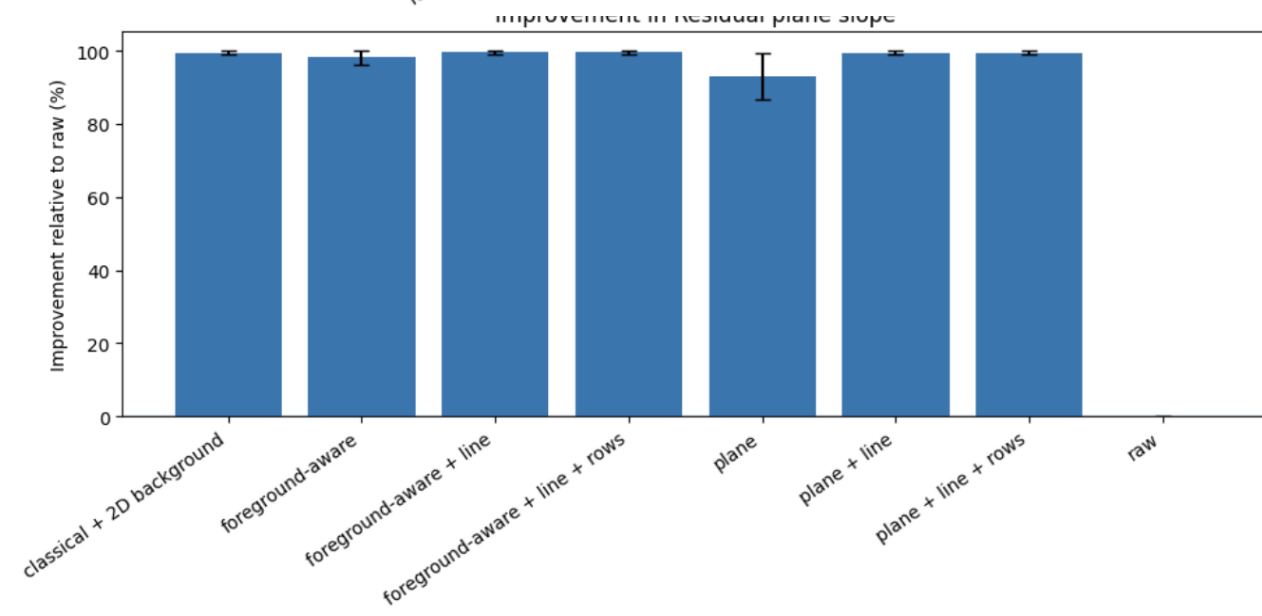
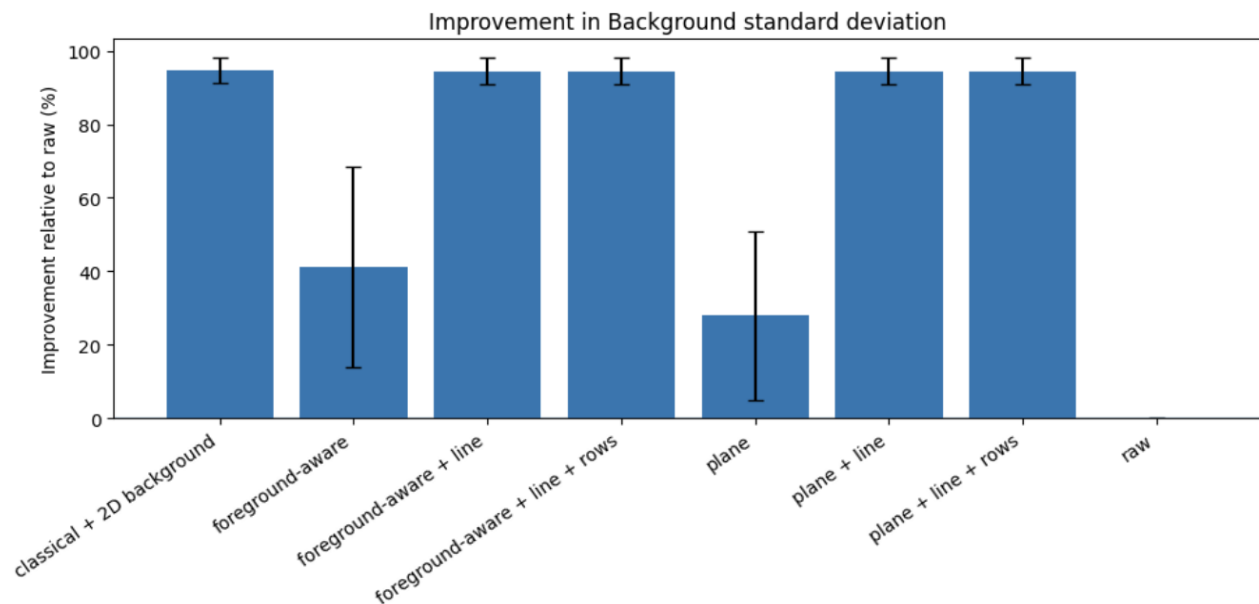
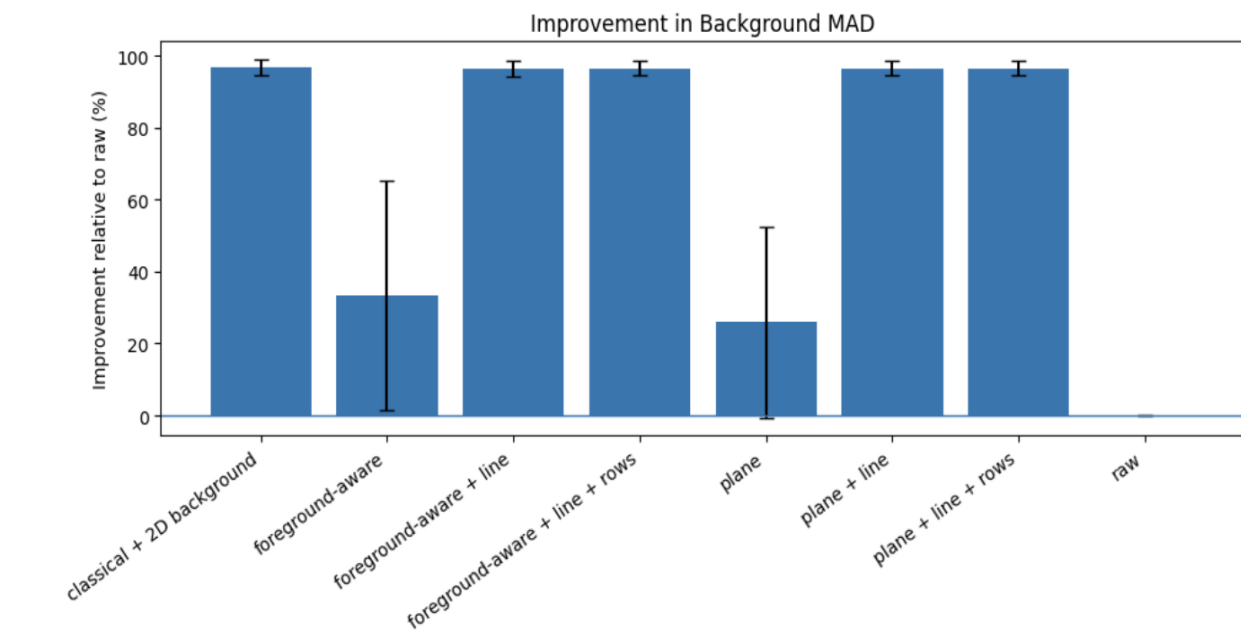


*U-net F1 score is affected by its threshold

3. Impact of pre-processing on detection F1 score

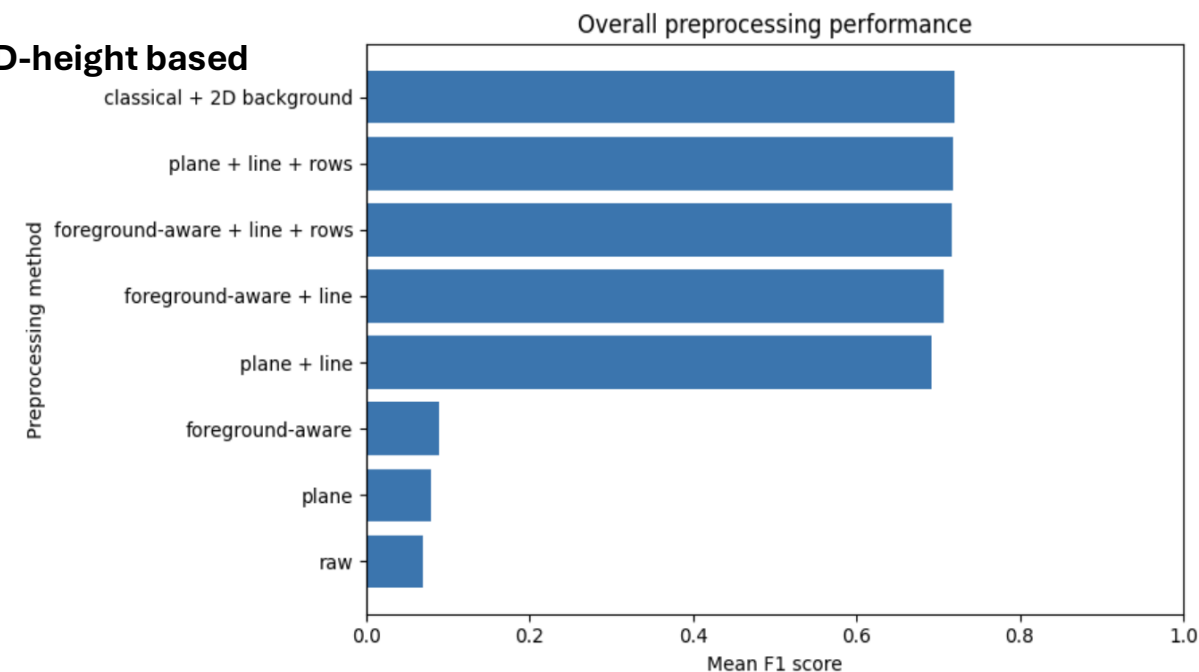


3.1 More detailed comparison ^{*Background-quality based}

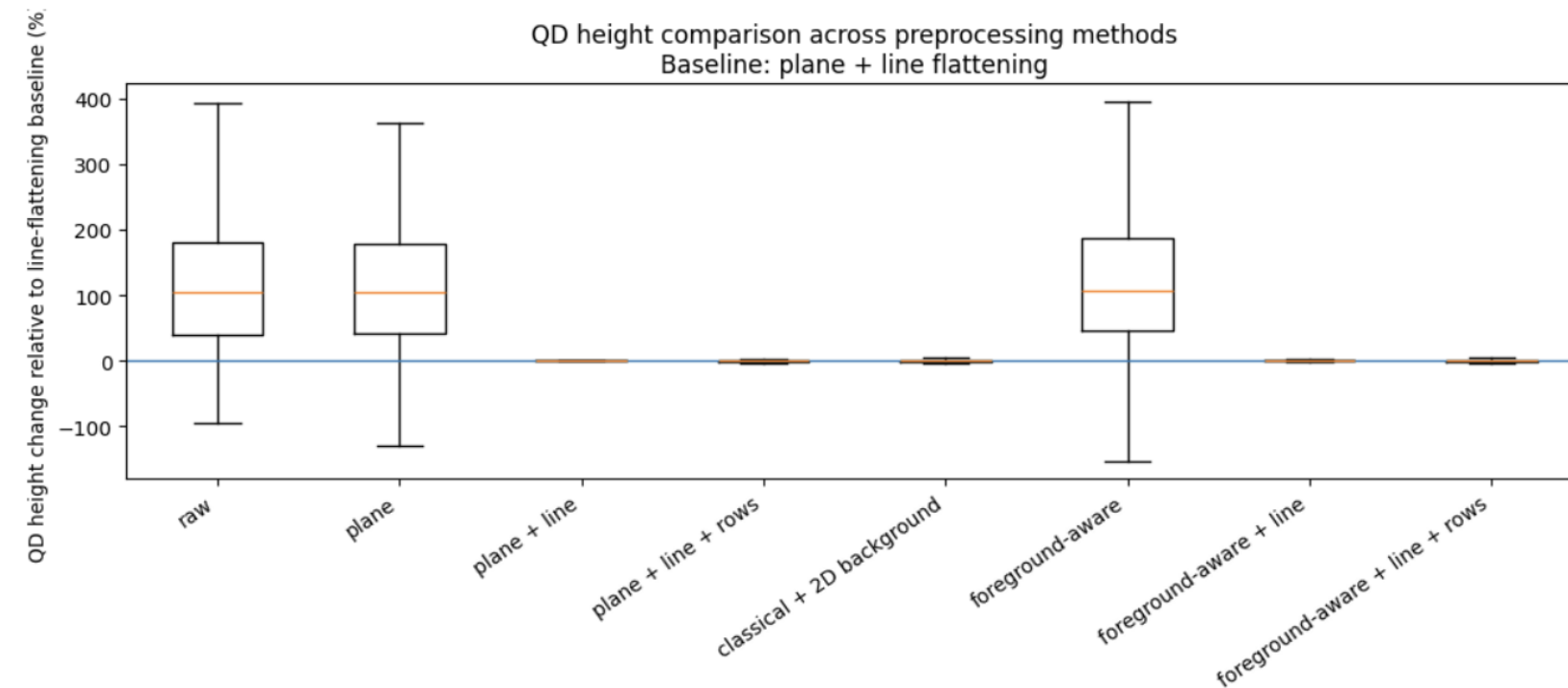


3.1 More detailed comparison

***QD-height based**

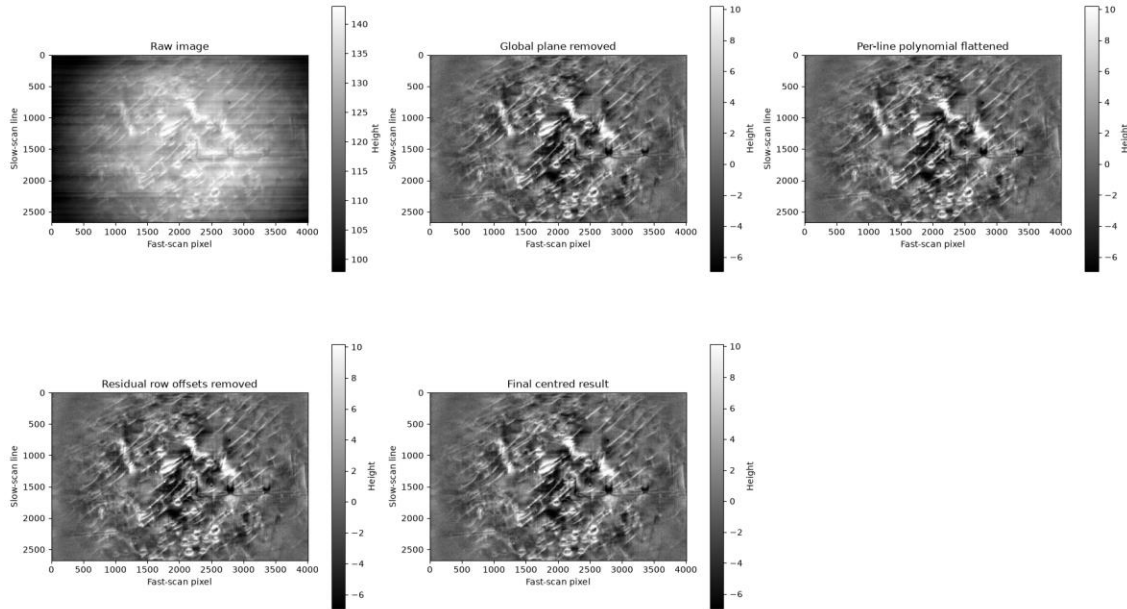


QD height comparison across preprocessing methods
Baseline: plane + line flattening



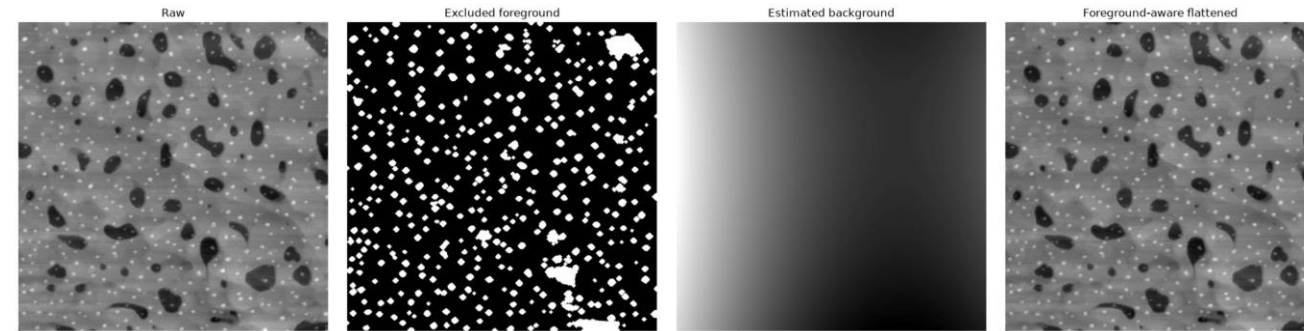
3.2 Methods specific to this project

1. Horizontal destriping (works even for diamonds!)



Works by smoothing the image differently in different directions

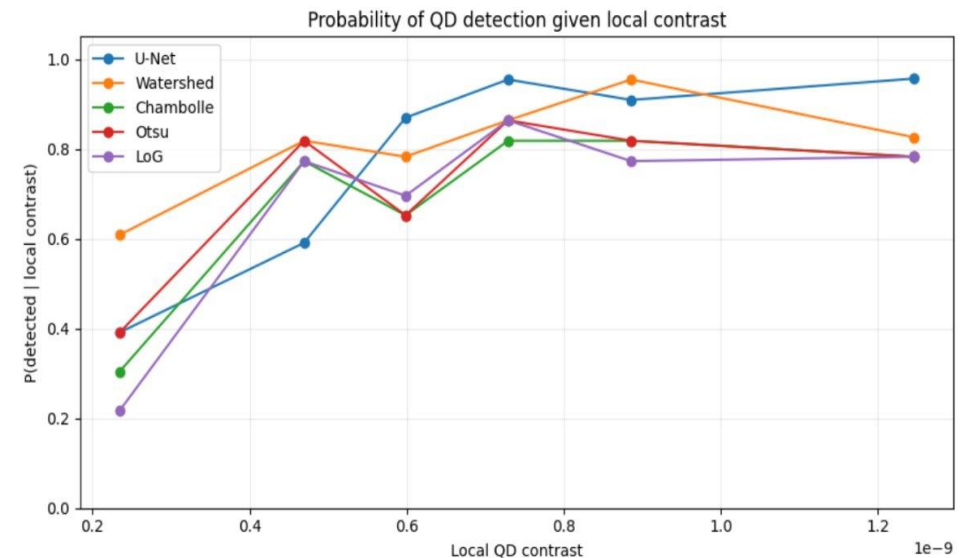
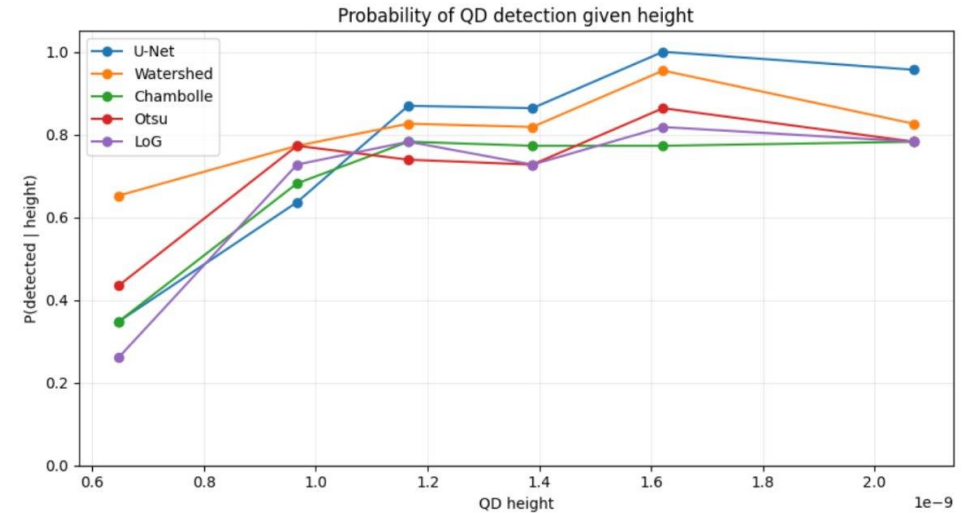
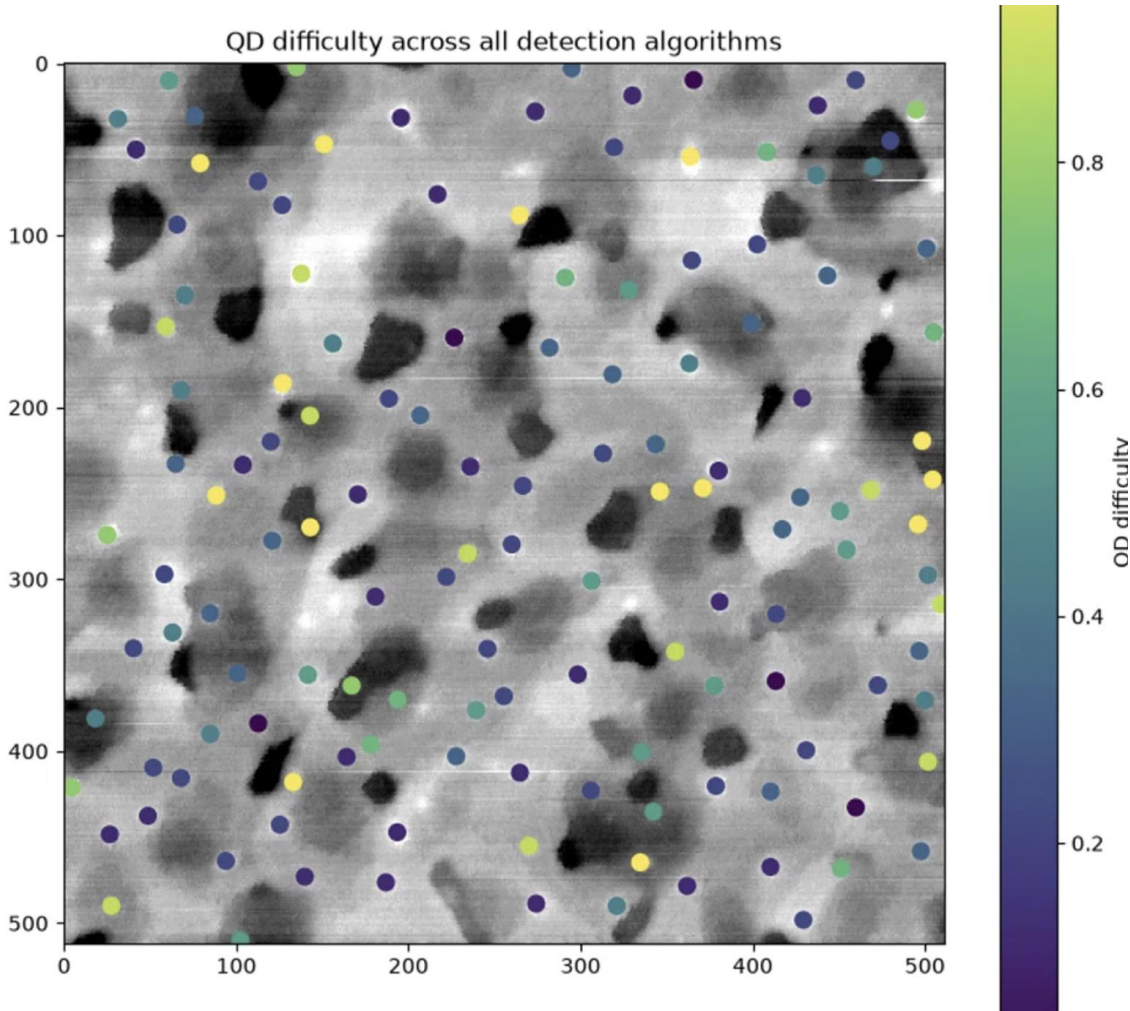
2. Foreground-aware pre-processing



Works by extracting the mask with the features affecting the background height and then smoothens the background without these features.

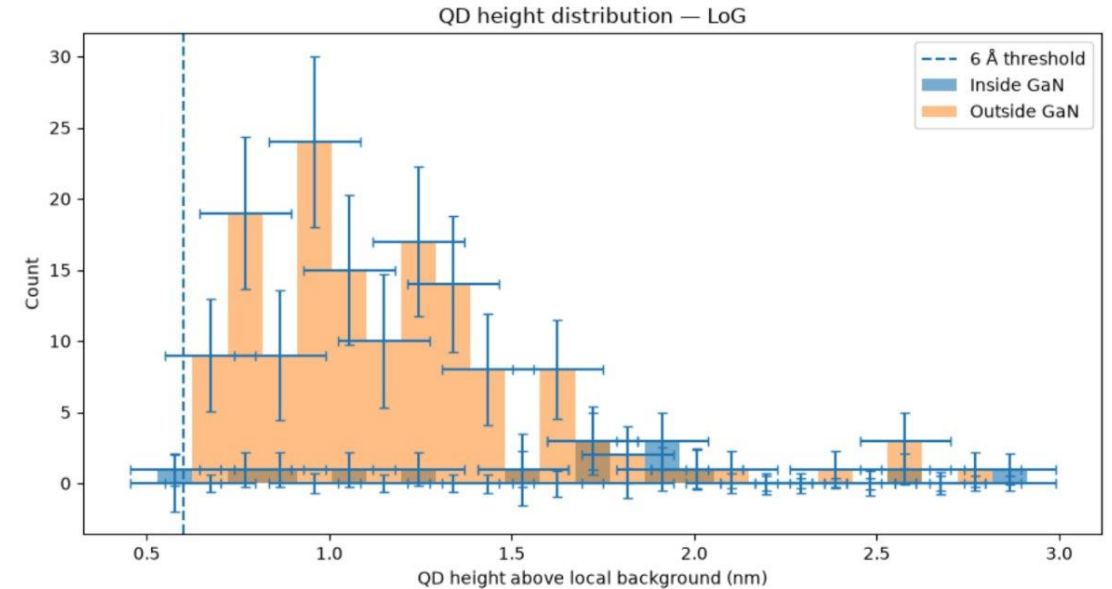
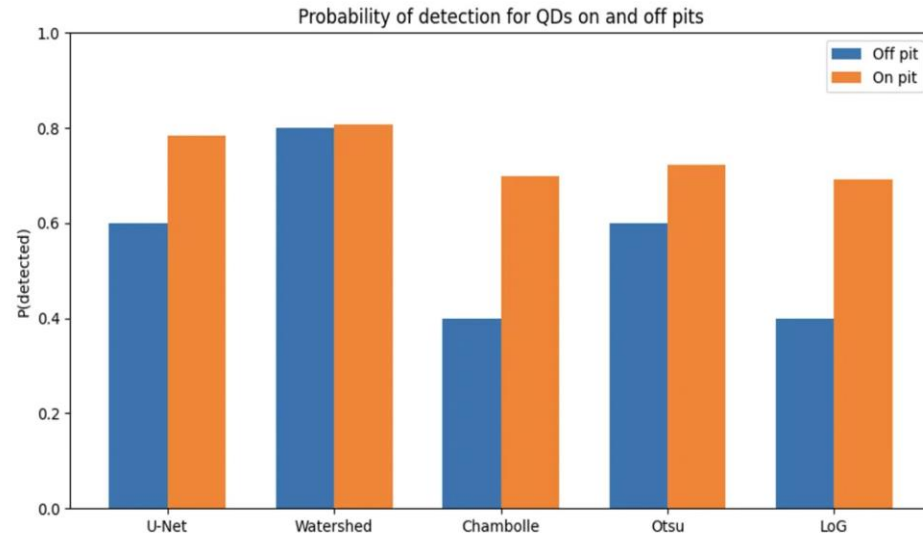
4. Why are certain dots hard to detect?

- **Physical Drivers:** Height and local contrast are top predictors.



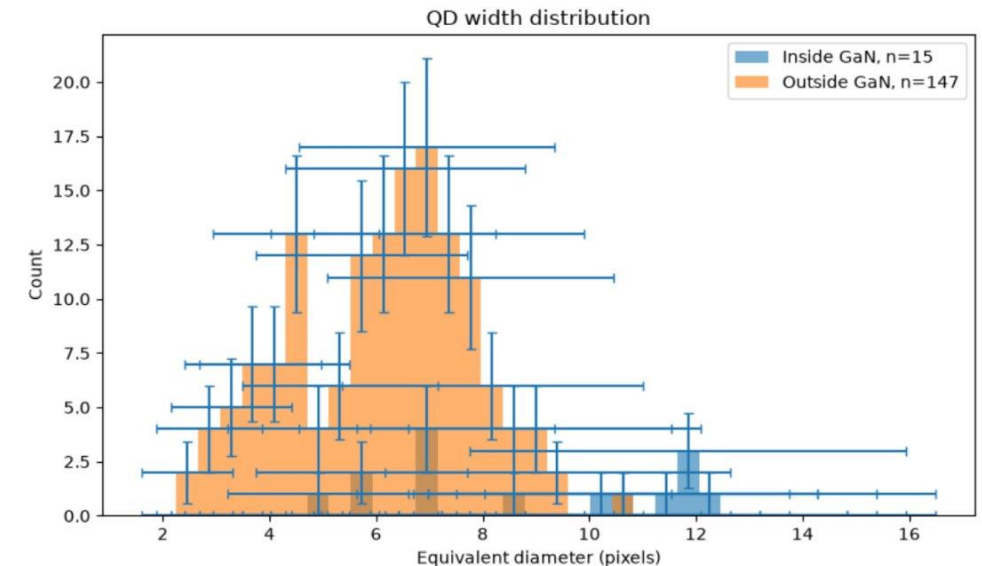
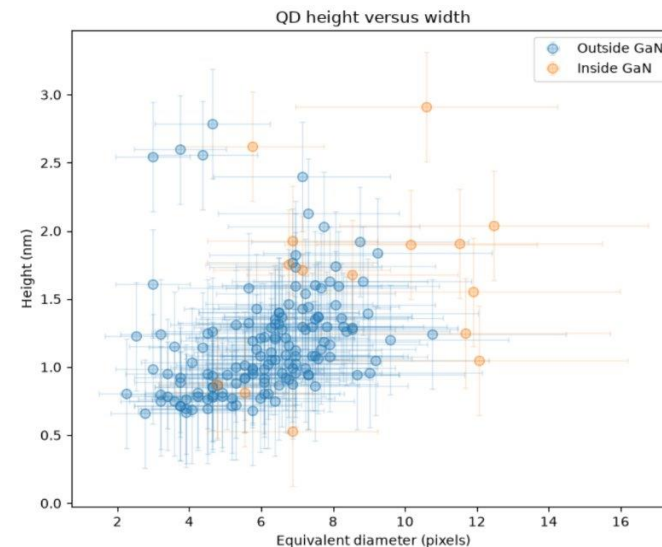
4. Why are certain dots hard to detect?

- **Spatial Difficulty:** Hard dots cluster in pits.

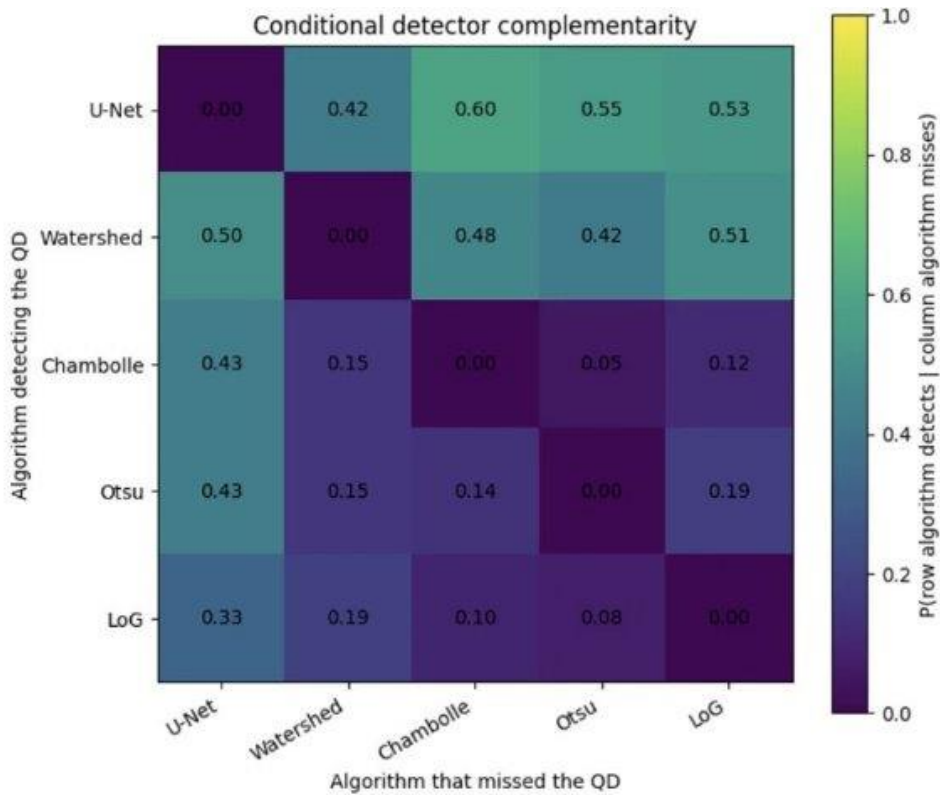


Uncertainty in R and h calculated using manually labelled Ground truth.

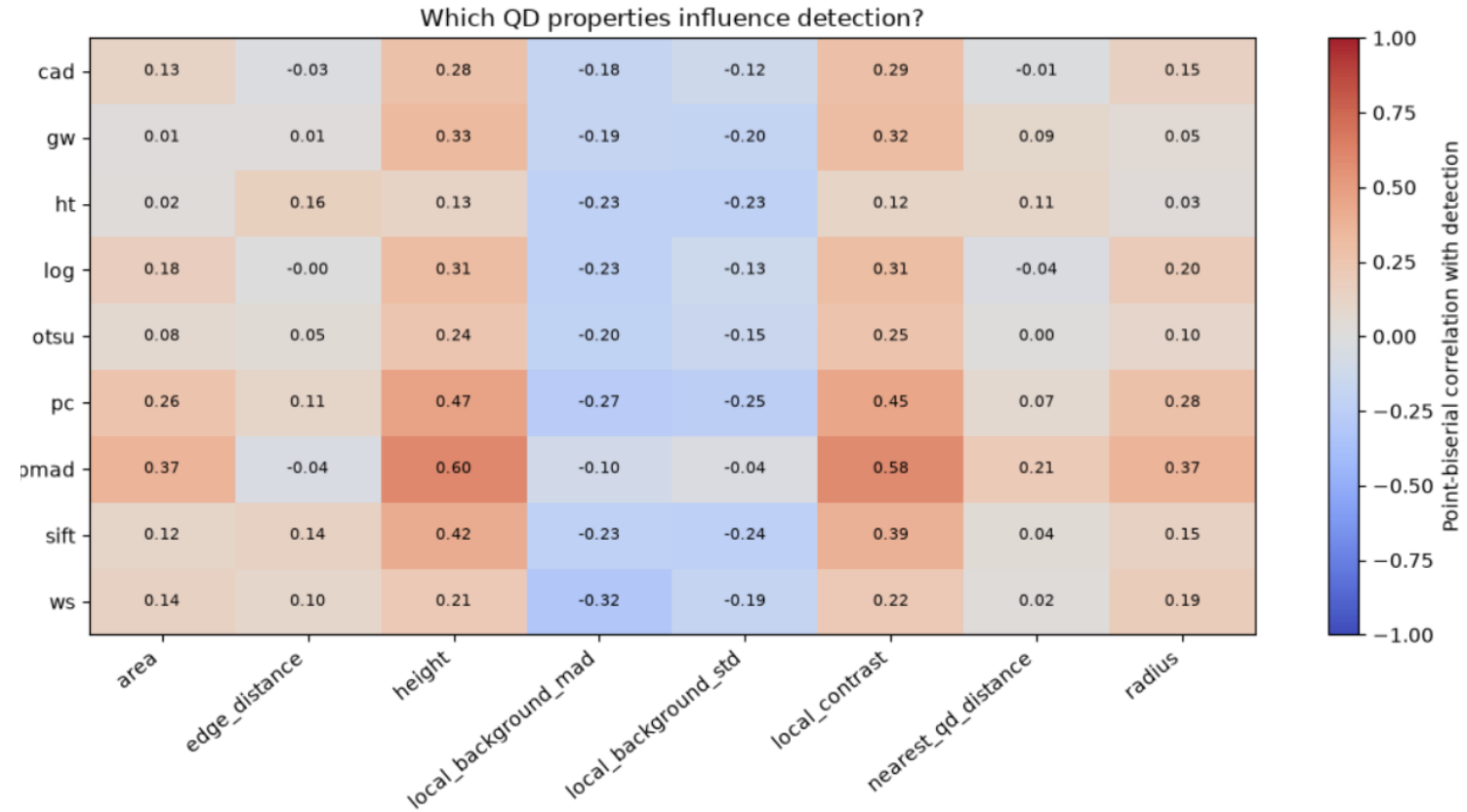
Uncertainty in count simulated using Monte Carlo.



4. Why are certain dots hard to detect?



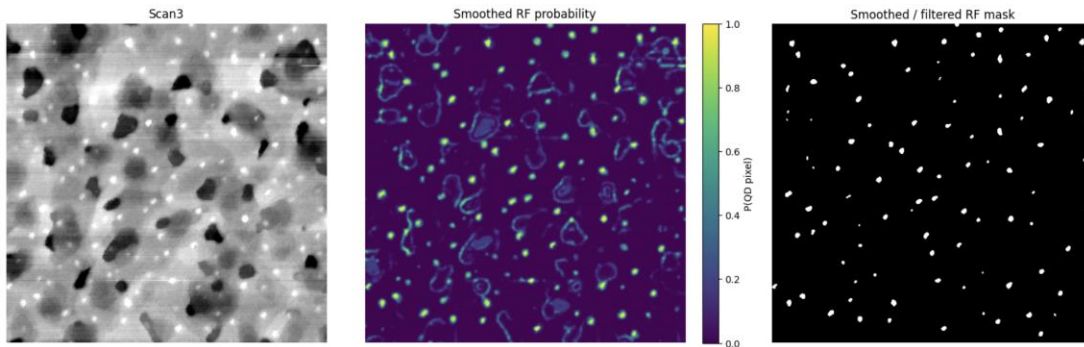
Different models have different probability of detecting the dots.



While other algorithms fail on pits, Watershed is the most robust. And is complementary to the U-net.

5. Random forests and Stacked Meta-Data Forests

1. Usual Random Forest, operating using conditional probabilities

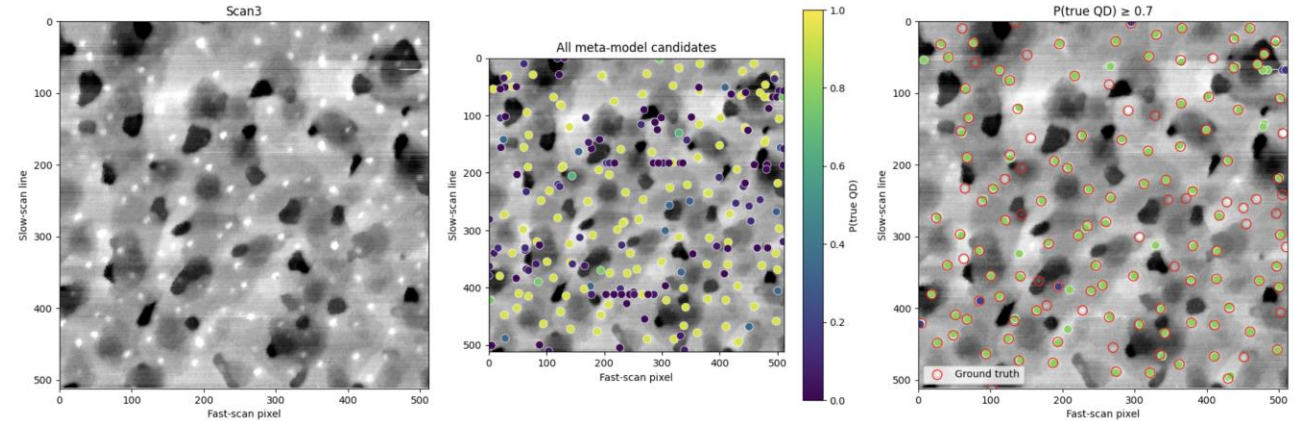


	TP	FP	FN	GT count	Predicted count	Precision	Recall	F1
Random Forest	83	14	55	138	97	0.85567	0.601449	0.706383

The random forest doesn't understand image structure by itself. It just sees the row of numbers.

one pixel → [height, smoothed height, LoG response, gradient, local mean, local std, ...]

2. Stacked forest using conditional probabilities of classical algorithms and the Unet when detecting a QD and using the complementarity matrix to verify this (Stacked Meta-Model)



Mean CV F1: 0.90

one pixel → [LoG response, Watershed response, U-net response, Otsu response, Chambolle responses]

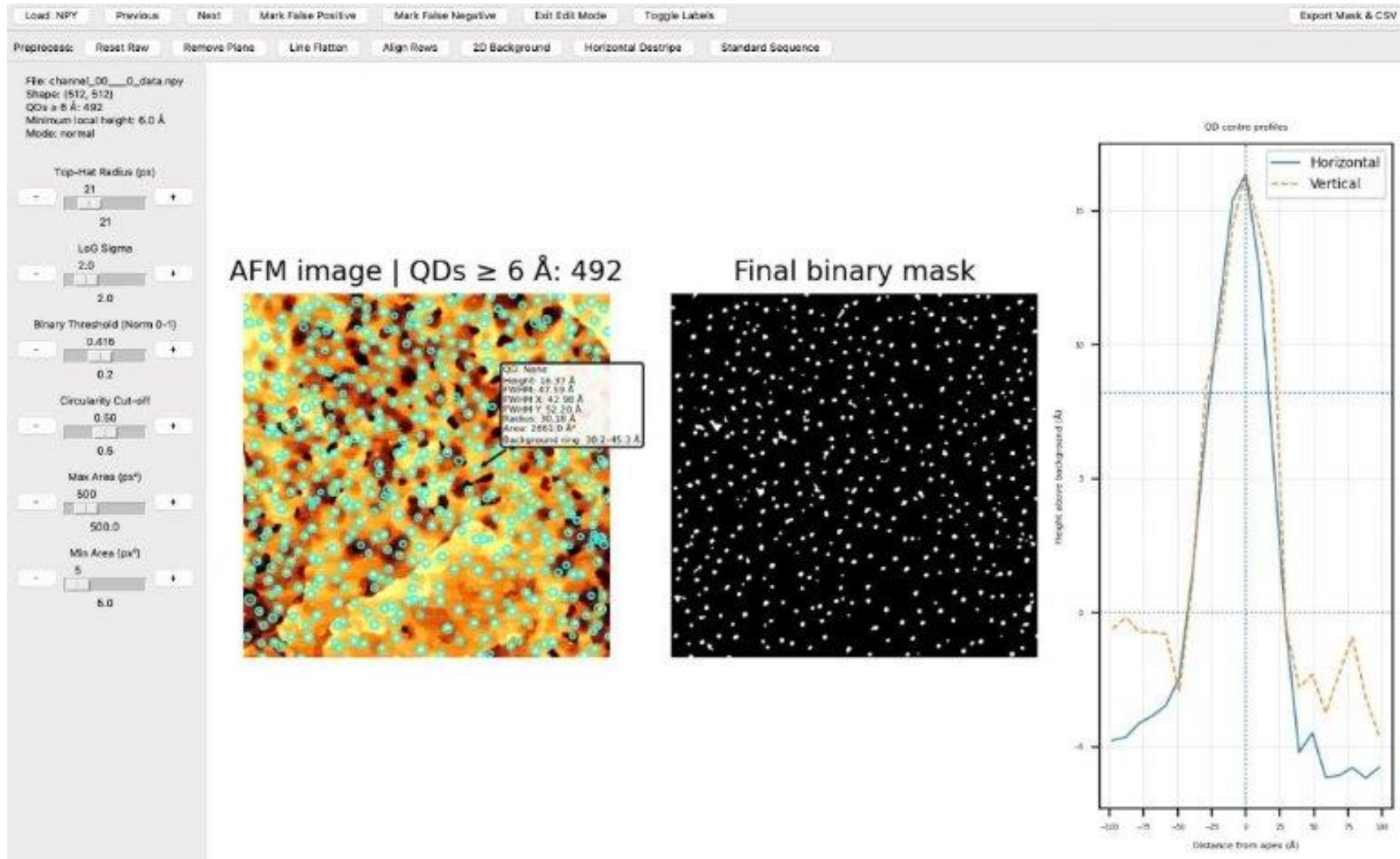
Aware of the conditional probabilities and algorithms' complementarity, this model makes its decisions.

Main GUI

- **Automatic QD detection**

- Manual correction of false positives/negatives

- **Instant QD measurements:** height, FWHM, area
- X/Y profile analysis relative to local background



GAN QUANTUM DOT ANALYSIS

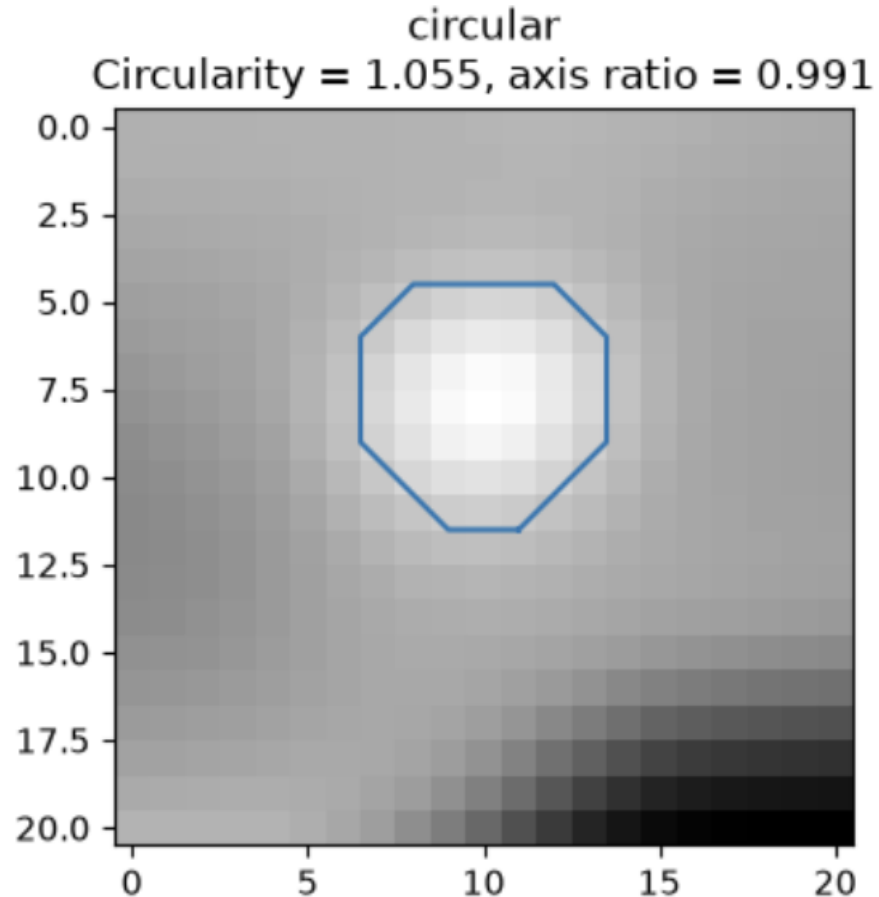
- > __pycache__
- > biology
- ▼ Data-Conversion
 - > Data-DISPLAY
 - > GWY
 - > NPY
 - > SPM
 - 🔗 gwy-to-npy.py
 - 📄 README.md
 - 🔗 spm-to-gwy.py
- ▼ Detection
 - > __pycache__
 - > Binary-Unet
 - > Biology-Inspired
 - > Classical-Optimised
 - > Random-Forest
 - > Stacked-Meta-Model
 - 📄 npy-data-QD-DETECTION.ipynb
 - 📄 npy-data-TRANSFORMS.ipynb
 - 📄 README.md
- ▼ Ground-Truth
 - > __pycache__
 - > Height-HWFM
 - > Images
 - > NPY-Ground-Truth
 - 🔗 QD-gui-npy-TRAINING-SETS.py
 - 📄 README.md

- ▼ Image-Preprocessing
 - > __pycache__
 - > Image-Destriping-Example
 - 🔗 basic_data_preprocessing.py
 - 🔗 foreground_preprocessing.py
 - 🔗 preprocessing_metrics.py
 - 📄 preprocessing-COMPARISON.ipynb
 - 📄 README.md
 - > Methodology
- ▼ QD-Analysis
 - > __pycache__
 - 📄 pit_distribution_analysis.ipynb
 - 🔗 QD_analysis.py
 - 📄 qd_conditional_detection.ipynb
 - 📄 qd_detection_failure_analysis.ipynb
 - 📄 README.md
- > venv-afm
- ▼ venv-ml
 - > bin
 - > etc
 - > include
 - > lib
 - > share
 - ⚙️ pyvenv.cfg
 - 📄 .gitignore
 - 🔗 QD-gui.py
 - 📄 README.md

Code structure changes

And every major folder now has a readme.

Conclusions and Future Work



- Image pre-processing is very important
- Stacked Random Forest can obtain an F1 score of 0.9! Is this better than my eyes?
- Tool to automate detection.

In the future:

- Add QD properties such as height, contrast and pit location to the meta-model.
- Use detector complementarity to design a difficulty-aware/custom U-Net loss.
- Integrate the final detector into the QD analysis GUI for automated detection and measurement.

*Dot with a funny circularity

((Detector doesn't account for the "halo" effect due to its small size, hence circularity happens to be larger than 1))